DESIGN SYSTEMS

TECHNICAL DATA SHEET SILIKA LT

CONSOLIDATING AND WATER-REPELLENT TREATMENT BASED ON LITHIUM SILICATES

DESCRIPTION

SILIKA LT is an aqueous solution based on modified lithium silicates, used for the surface treatment of cement-based substrates, indoors and outdoors, horizontal and vertical. Liquid surface treatment with a consolidating, water-repellent and protective effect for new or existing concrete floors and concrete surfaces finished with a dry-shake mineral hardener.

SILIKA LT penetrates the surface pores of the substrate and reacts with the free lime that is normally produced during the hydration phase of the cement. This reaction produces highly resistant, insoluble hydrated calcium silicates, which increase the compactness and strength of the cementitious matrix. The treated surface takes on a glossier appearance. The treated surfaces are characterised by:

- Increased wear resistance and chemical resistance.
- Reduced dust formation.
- Excellent reduction in the absorption of water and liquids in general, and of the salts dissolved in them, such as chlorides, which are harmful to any metal reinforcement present.
- Water vapour permeability maintained.
- Resistance to weathering and UV rays.
- Reduced dirt retention and formation of stains and efflorescence.

Advantages: ease of use and application; the treated surface is easy to clean and maintain; it improves the appearance of the surface, making it glossier and protecting it from the formation of stains and efflorescence; it increases the durability of the concrete floor; it reduces the alkali-aggregate reaction phenomenon (pop-out) in the presence of alkali-reactive aggregates in the substrate; it protects against damage from de-icing salts, the action of weathering and freeze-thaw cycles.

USES

SILIKA LT is suitable for new or existing cementitious floors, including those finished with a dry-shake mineral hardener, and is widely used in the industrial, residential and commercial sectors: hangars, garages, car parks; shopping centres, areas open to the public; workshops, engineering companies, manufacturing industries in general; warehouses, logistics facilities, outdoor yards; craft workshops; restaurants, schools, museums, etc. It can also be used in the final stages of the polishing treatment in the case of polished concrete floors.

SUBSTRATE PREPARATION

The cementitious substrate to be treated must be perfectly clean and dry, free from contaminants, previous protective or curing-compound treatments, film-forming resin treatments, dust, friable or detaching parts. Do not use acidic-pH detergents for any preliminary washing or cleaning of the surface. If the surface has

already been treated with acidic detergents or products, for example pigments, carry out a pH-neutralising wash before applying the product.

APPLICATION

SILIKA LT is ready to use and only needs to be stirred before use. It must be applied to existing or newly laid concrete floors, 1-2 days after casting and in any case after the joints have been cut, by spraying it evenly with a low-pressure pump onto the surface to be treated or by pouring it directly onto the floor and spreading it immediately and evenly.

The treated surface must be kept continuously wet with SILIKA LT for at least 20-30 minutes: it is very important to prevent dry areas from forming, in order to allow correct and uniform penetration of the product. At the end of this period, remove the excess liquid, drying the treated surface. Once these operations are completed, it is advisable not to walk on the surface for at least 24 hours.

The equipment used for preparation and application can be cleaned simply with soap and water. Once the product has hardened, it can only be removed mechanically.

WARNINGS

- Do not apply SILIKA LT on very dusty and/or friable or weak substrates.
- Do not apply SILIKA LT on dirty surfaces contaminated by oils and/or greasy substances.
- Before applying SILIKA LT, check that the surface is uniformly dry, without residual moisture or standing water.
- Always carry out a preliminary test on a small portion of the surface to check for any contaminants that are not visible and/or not compatible.
- If resin products are to be applied subsequently to the treated surface, carry out adequate mechanical preparation of the surface beforehand, before laying the resin system, and carry out preliminary tests on a small portion of the surface to check adhesion and compatibility.

For safe use of the product, refer to the latest version of the safety data sheet. Product for professional use.

PRODUCT DATA

DESCRIPTION	RESULT
Consistency	Liquid
Colour	Transparent
Density	1,07 g/cm^3
pH value (ISO 4316)	11,5
Consumption	0,05-0,30 kg/m^2, depending on the type of substrate, its absorbency and the ambient conditions during application
Application	By spray with a low-pressure pump, or poured and spread directly

Trafficability	At least 24 hours after treatment
Tool cleaning	Soap and water (before hardening)
Packaging	25 kg can
Storage	12 months in the original sealed packaging, in a dry place at a temperature between +5°C and +40°C. Sensitive to frost.

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.